



NodeJS Introduction



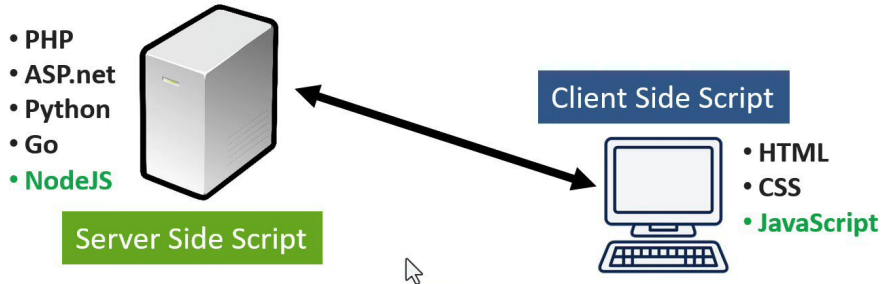
Node.js is an open-source, cross-platform
JavaScript runtime environment that allows you to
run JavaScript code outside a web browser.



NodeJS Introduction



Node.js is an open-source, cross-platform **JavaScript runtime environment** that allows you to run JavaScript code outside a web browser.

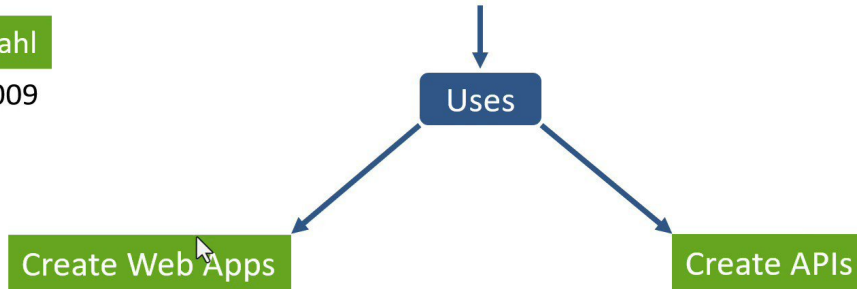




Node.js is an open-source, cross-platform **JavaScript runtime environment** that allows you to run JavaScript code outside a web browser.

Ryan Dahl

May 2009





Node.js is an open-source, cross-platform **JavaScript runtime environment** that allows you to run JavaScript code outside a web browser.

Ryan Dahl

May 2009

Chrome's V8 JavaScript engine

Uses

Create Web Apps

Create APIs

- Faster
- Lightweight
- Efficient
- Scalable



Use of NodeJS

- CMS (Content Management System)
- ERP(Enterprise Resource Planning)
- CRM(Customer Relationship Management)
- Mail server
- Real-time Applications : Chatting System, online games
- File Storage System
- Command-Line Tools
- Backend for Mobile Apps
- Streaming Data : Video streaming, Audio streaming, and processing sensor data from IoT devices.





Benefits of NodeJS

- **Performance:** V8 engine and non-blocking I/O make it very fast and efficient.
- **Scalability:** Easily handles a large number of concurrent connections.
- **Full-Stack JavaScript:** Developers can use JavaScript for both the front-end and back-end, which reduces context switching and improves development speed.
- **Large Ecosystem:** NPM provides a vast collection of libraries and tools.
- **Community:** A large and active community provides support, resources, and open-source contributions.
- **Real-time Capabilities:** Excellent for building real-time applications.
- **Easy to Learn:** If you already know JavaScript, the learning curve for Node.js is relatively gentle.
- **Cost-Effective:** Can often reduce infrastructure costs due to its efficient resource utilization.



Popular Companies using NodeJS

- Netflix
- LinkedIn
- Uber
- PayPal
- NASA
- Walmart
- Medium
- Trello
- eBay



Express.js is a fast, minimal, and flexible **web framework** for **Node.js** that simplifies building **web applications** and **APIs** in a more structured and organized way.





What is web framework ?

Web frameworks are sets of **pre-written code** and **libraries** that provide a foundation for developing software applications.

Pre-written Code & Library

- Tools
- Components
- Modules





What is web framework ?

Web frameworks are sets of **pre-written code** and **libraries** that provide a foundation for developing software applications.

Pre-written Code & Library

- Tools
- Components
- Modules

Examples

- Database Component
- Caching
- Pagination
- Session management
- Form Handling
- Security mechanisms
- User authentication
- APIs
- Payment Gateways 



What is web framework ?

Web frameworks are sets of **pre-written code** and **libraries** that provide a foundation for developing software applications.

Pre-written Code & Library

- Tools
- Components
- Modules

Examples

- Database Component
- Caching
- Pagination
- Session management
- Form Handling
- Security mechanisms
- User authentication
- APIs
- Payment Gateways

Benefits

- Code organization
- Reusability
- Standardization 
- Testing & debugging support
- Community and support



Type of Frameworks

Opinionated Framework

Has a defined architecture (MVC, MVVM, etc.).

-
- Ruby on Rails (**Ruby**)
 - Django (**Python**)
 - Spring Boot (**Java**)
 - Laravel (**PHP**)

Unopinionated Framework

No strict rules or predefined workflow.

-
- Express.js (**Node.js**)
 - Flask (**Python**)
 - FastAPI (**Python**)
 - Koa.js (**Node.js**)
 - React.js



What we will learn in this series

- Parameters
- Requests
- APIs
- Template Engine
- Middleware
- Validation
- File Upload
- Routers
- Cookies
- Sessions
- Authentication
- Error Handling
- Testing
- Events
- Streams
- Sockets



VS Code Extensions

- ExpressJs 4 Snippets – ([Güray Yazar](#))
- ExpressJS Snippets – ([Xpertpk](#))
- MERN Snippets – ([mohamed-khaled](#))
- EJS language support – ([DigitalBrainstem](#))
- MongoDB for VS Code – ([MongoDB](#))
- Path Intellisense – ([Christian Kohler](#))
- ESLint – ([Microsoft](#))



npm init -y



Package.json

- Name
- Version
- Author
- Description
- Keywords





npm init -y

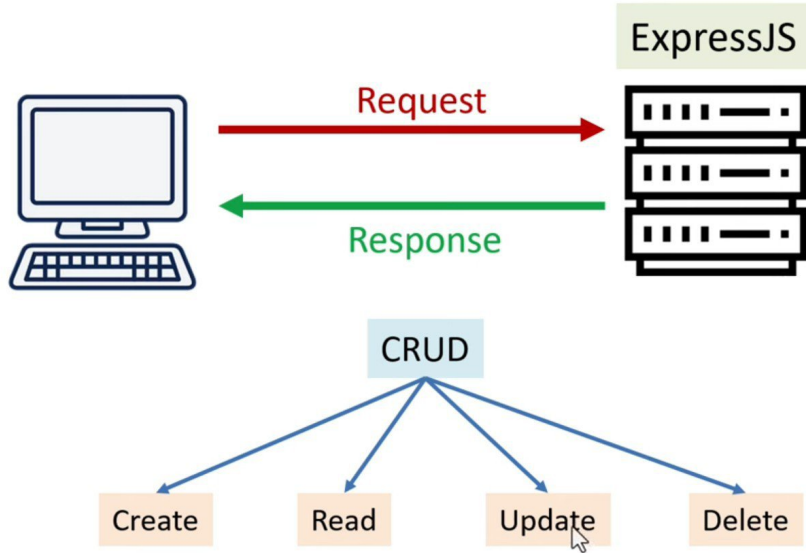


Package.json

- Name
- Version
- Author
- Description
- Keywords
- Packages (Dependencies)



How ExpressJS Works ?





How to use ExpressJS

1

```
const express = require('express')  
const app = express()
```

2

```
app.listen(3000, () => {  
  console.log(`Example app listening on port 3000`)  
})
```

HTTP requests

http://localhost:3000/



http://localhost:3000/about

http://localhost:3000/gallery



How to use ExpressJS

1 `const express = require('express')`
`const app = express()`

2 `app.listen(3000, () => {`
 `console.log(`Example app listening on port 3000`)`
`})`

HTTP requests

`http://localhost:3000/` 
`http://localhost:3000/about`
`http://localhost:3000/gallery`

```
app.get('/', (req, res) => {  
  res.send('Hello !')  
})
```



Route



URL

- Webpage URL
- APIs URL

HTML

`http://localhost/about.html`

`http://localhost/pages/about.html`



ExpressJS Route

`http://localhost:3000/`

`http://localhost:3000/about`  `abc.html`

`http://localhost:3000/about/user` Nested Route

`http://localhost:3000/user/10` Route Parameters

`http://localhost:3000/user/yahubaba/rollno/12`

`http://localhost:3000/search?name=yahubaba`




ExpressJS Route

`http://localhost:3000/`

`http://localhost:3000/about`  `abc.html`

`http://localhost:3000/about/user` Nested Route

`http://localhost:3000/user/10` Route Parameters

`http://localhost:3000/user/yahubaba/rollno/12`

`http://localhost:3000/search?name=yahubaba` Query Parameters



When to use Route ?

Express **JS**

HTTP Request

Read

```
app.get('/about', (req, res) => {  
  })
```

Create

```
app.post('/about', (req, res) => {  
  })
```

Update

```
app.put('/about', (req, res) => {  
  })
```

Delete

```
app.delete('/about', (req, res) => {  
  })
```



Pass Value through Route Parameters

http://localhost:3000/**user**/**10** 

```
app.get('/user/:userid', (req, res) => {  
  res.send(req.params)  
})
```



EXPLORER



index.js



FIRST-PROJECT

index.js > ...



> node_modules

index.js

package-lock.json

package.json



```
6  })
7
8  app.get('/', (req, res) => {
9    res.send("<h1>Welcome to Home Page</h1>")
10 })
11
12 app.get('/user/:userid/book/:bookid', (req, res) => {
13   res.send(req.params)
14 })
15
16
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



EXPLORER



index.js x



FIRST-PROJECT

index.js > ...



> node_modules

index.js

package-lock.json

package.json



```
6  })
7
8  app.get('/', (req, res) => {
9    res.send("<h1>Welcome to Home Page</h1>")
10 })
11
12 app.get('/user/:userid-:bookid', (req, res) => {
13   res.send("Book Id : " + req.params.bookid)
14 })
15
16 |
```

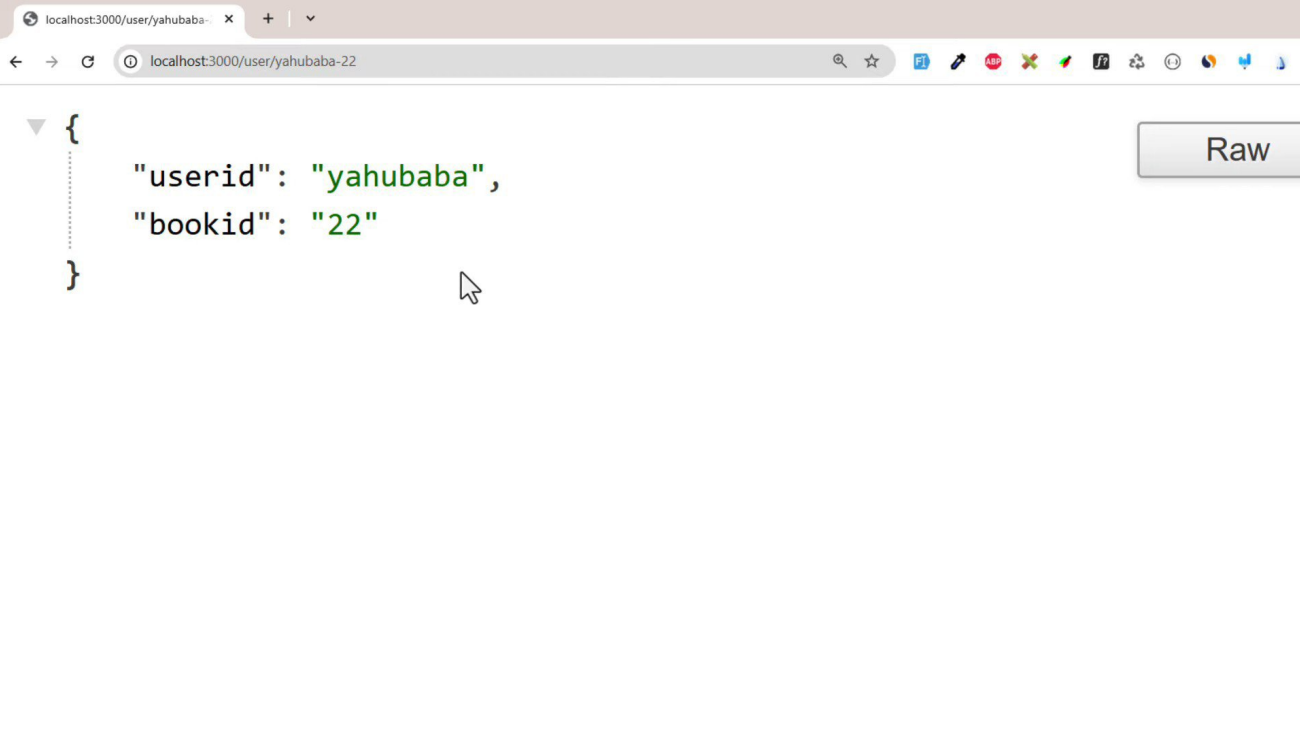
PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS





EXPLORER



index.js



FIRST-PROJECT

> node_modules

index.js

package-lock.json

package.json



index.js > app.get('/search') callback

```
9   res.send("<h1>Welcome to Home Page</h1>")
10 }
11
12 app.get('/user/:userid-:bookid', (req, res) => {
13   res.send(req.params)
14 })
15
16 app.get('/search', (req, res) => {
17   res.send(req.query)
18 })
```

I

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS


```
▼ {  
  "name": "yahubaba",  
  "age": "20",  
  "city": "chandigarh"  
}
```



EXPLORER



index.js



FIRST-PROJECT

> node_modules

index.js

package-lock.json

package.json



index.js > app.get('/search') callback

```
9   res.send("<h1>Welcome to Home Page</h1>")
10 }
11
12 app.get('/user/:userid-:bookid', (req, res) => {
13   res.send(req.params)
14 })
15
16 app.get('/search', (req, res) => {
17   const name = req.query.name
18   const age = req.query.age
19
20   res.send(`Search results for Name : ${name} , Age : ${age}`)
21 })
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



Response Methods

YahuBaba

www.Yahubaba.com



ExpressJS Response Methods

- `res.send()` → Text, HTML, Object, Array, Buffers
- `res.json()` → JSON (JavaScript Object Notation)
- `res.jsonp()` → JSONP (JSON with Padding)
- `res.redirect()`
- `res.render()` → Open HTML File using a template engine
- `res.download()` → PDF, Image File, Music, Video, Excel etc..
- `res.sendFile()`
- `res.end()`
- `res.sendStatus()`
- `res.headersSent()`
- `res.set()`
- `res.get()`



Redirect Code

Code	Description
301	A permanent redirect used when a resource has been moved to a new URL.
302	A temporary redirect used when a resource has been temporarily moved to a new URL.
303	A temporary redirect used after a POST or PUT operation.
307	A temporary redirect similar to 302, but better for sites with non-GET operations.
308	The permanent counterpart to the 307 redirect.



EXPLORER



index.js



FIRST-PROJECT



node_modules



index.js

package-lock.json

package.json



```
8 app.get('/', (req, res) => {
9
10   res.jsonp({ name: 'YahuBaba', age: 25})
11 })
12
13 app.get('/about', (req, res) => {
14   res.redirect(301, 'https://www.google.com');
15 })
16
17 app.get('/user', (req, res) => {
18   res.send("<h1>User Page</h1>")
19 })
20
21
```



EXPLORER



index.js × user.ejs



FIRST-PR... index.js > app.get('/user') callback



> node_modules



views



user.ejs



index.js C:\YahooBaba\Learn-NodeJS\first-project\views\user.ejs



package-lock.json



package.json



14

15

16

17

18

19

20

21

22

23

```
app.get('/about', (req, res) => {
```

```
  res.redirect('..');
```

```
})
```

```
app.get('/user', (req, res) => {
```

```
  res.render('user')
```

```
})
```



EXPLORER



index.js × user.ejs



FIRST-PROJECT

index.js > ...



> node_modules

views

user.ejs

index.js

package-lock.json

package.json

```
4 app.listen(3000, ()=>{
5   console.log('Successfully Connected on port 3000.')
6 })
7 
8 app.set('view engine', 'ejs')
9 
10 app.get('/', (req, res) => {
11   res.jsonp({ name: 'YahuBaba', age: 25})
12 })
13 
14 
15 app.get('/about', (req, res) => {
16   res.redirect('..');
17 })
18 
19 app.get('/user', (req, res) => {
20   res.render('user')
21 })
22 
23
```




EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/download') callback

14

15

16

17

18

19

20

21

22

23

`app.get('/about', (req, res) => {` `res.redirect('..');``})``app.get('/download', (req, res) => {` `res.download('./files/shortcuts.pdf', 'Document.pdf')``})`

22

23

I



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/download') callback

14

15 app.get('/about', (req, res) => {

16 | res.redirect('..');

17 | })

18

19 app.get('/download', (req, res) => {

20 | ⚠ res.sendFile('./files/shortcuts.pdf', 'Document.pdf')

21 | })

22

23



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/download') callback

14

15 app.get('/about', (req, res) => {

16 res.redirect('..');

17 })

18

19 app.get('/download', (req, res) => {

20 res.sendFile('./files/shortcuts.pdf')

21 })

22

23



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

14

15 app.get('/about', (req, res) => {

16 res.redirect('..');

17 })

18

19 app.get('/download', (req, res) => {

20 res.sendFile(__dirname + '/files/shortcuts.pdf')

21 })

22

23



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

14

15 app.get('/about', (req, res) => {

16 | res.redirect('..');

17 | })

18

19 app.get('/error', (req, res) => {

20 | res.sendStatus(200)

21 | })

22

23



Server Status Codes

200 - OK

201 - Created

403 - Forbidden

404 - Not Found

500 - Internal Server Error

503 - Service Unavailable

504 - Gateway Timeout



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/error') callback

```
14
15 app.get('/about', (req, res) => {
16   res.redirect('..');
17 })
18
19 app.get('/error', (req, res) => {
20   res.status(200).send("Hello")
21 })
22
23
```



EXPLORER



index.js



user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/check') callback

```
17 })  
18  
19 app.get('/check', (req, res) => {  
20   console.log(res.headersSent)  
21   res.send("Hello")  
22   console.log(res.headersSent)  
23 })  
24
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
[nodemon] restarting due to changes...  
[nodemon] starting `node index.js`  
Successfully Connected on port 3000.  
false  
true  
[
```




EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

17 })

18

19 app.get('/check', (req, res) => {

20 res.set('custom-header', 'hello123')

21 console.log(res.get('custom-header'))

22 res.send("Header Set")

23 })

24

25

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

[nodemon] restarting due to changes...



EXPLORER



index.js × user.ejs



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

17

})

18



19

app.get('/check', (req, res) => {

20

res.set('custom-header', 'hello123')

21

console.log(res.get('custom-header'))

22

res.send("Header Set")

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
Successfully Connected on port 3000.  
[nodemon] restarting due to changes...  
[nodemon] starting `node index.js`  
Successfully Connected on port 3000.  
hello123  
█
```



Request Properties & Methods

YahuBaba

www.Yahubaba.com



ExpressJS Request Properties & Methods

Properties

- req.params
- req.query
- req.body
- req.cookies
- req.hostname
- req.ip
- req.ips
- req.method
- req.originalUrl
- req.path
- req.protocol
- req.secure
- req.route

JSON, form data

Methods

- req.accepts HTML, JSON, XML
- req.get
- req.is JSON, form data or plain text
- req.range





EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

```
1  const express = require('express')
2  const app = express()
3
4  app.listen(3000, ()=>{
5    console.log('Successfully Connected on port 3000.')
6  })
7
8  app.set('view engine', 'ejs')
9
10 app.use(express.json())
11
12 app.get('/', (req, res) => {
13
14   res.jsonp({ name: 'YahuBaba', age: 25})
15 })
16
17 app.get('/about', (req, res) => {
18
19 })
20
21
```



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > app.post('/about') callback

```
12 app.get('/', (req, res) => {
13   |
14   |   res.jsonp({ name: 'YahuBaba', age: 25})
15   | })
16
17 app.post('/about', (req, res) => {
18   res.send(req.body)
19 })
20
21
22
```

Import

Overview

POST localhost:3000/about



No environment



HTTP

localhost:3000/about

Save



Share



POST



localhost:3000/about



Send



Body



Cookies

Response

History



raw



JSON



Beautify

```
1 {  
2   "name": "Yahubaba",  
3   "age": 25  
4 }
```



Click Send to get a response

Import

Overview

POST localhost:3000/about

+

Dropdown

No environment

Dropdown

Help

HTTP

localhost:3000/about

Save

Dropdown

Share

</>

POST

Dropdown

localhost:3000/about

Send

Dropdown

Refresh

Body

Dropdown

Cookies

Body

Dropdown

Refresh

200 OK

•

2.65 s

•

263 B

•

Globe

...

raw

Dropdown

JSON

Dropdown

Beautify

{ } JSON

Dropdown

Preview

Visualize

Dropdown

Dropdown

Copy

Copy

Search

Link

```
1 {
2   "name": "Yahubaba",
3   "age": 25
4 }
```

```
1 {
2   "name": "Yahubaba",
3   "age": 25
4 }
```




EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

```
9
10 app.use(express.json())
11
12 app.get('/', (req, res) => {
13
14   res.jsonp({ name: 'YahuBaba', age: 25})
15 })
16
17 app.post('/about', (req, res) => {
18   res.send(req.body)
19 })
20
21
22
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

EXPLORER index.js

FIRST-PROJECT index.js > extended

- > files
- > node_modules
- > views
 - user.ejs
 - index.js
- package-lock.json
- package.json

```
4 app.listen(3000, ()=>{
5   console.log('Successfully Connected on port 3000')
6 })
7
8 app.set('view engine', 'ejs')
9
10 app.use(express.json())
11 app.use(express.urlencoded({extended: false}))
12
13 app.get('/', (req, res) => {
14
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.post('/about', (req, res) => {
19   res.send(req.body)
20 })
21
22
23
```

`urlencoded(options?:
bodyParser.OptionsUrlencoded):
createServer.NextHandleFunction`

Returns middleware that only parses urlencoded bodies and only looks at requests where the Content-Type header matches the type option

Report

Overview

POST localhost:3000/about

+

Dropdown

No environment

Dropdown

Help

...

HTTP

localhost:3000/about

Save

Dropdown

Share

</>

POST

Dropdown

localhost:3000/about

Send

Dropdown

Refresh

Body

Dropdown

Cookies

x-www-form-urlencoded

	Key	Value	Description	...	Bulk Edit
☒	name	Akshay Kumar ↗ ...			
☒	age	25			
	Key	Value	Description		

Body

Dropdown

Refresh

200 OK

•

141 ms

•

271 B

•

Globe

...

{ } JSON

Preview

Visualize

Dropdown

List

Copy

Search

Link

1

{

2

"name": "Akshay Kumar\n",

3

"age": "25"

4

}

Report

Overview

POST localhost:3000/about



No environment



...



localhost:3000/about



Share



POST



localhost:3000/about

Send



Body



Cookies

x-www-form-urlencoded

	Key	Value	Description	...	Bulk Edit
☒	name	Akshay Kumar ↗ ...			
☒	age	25			
☒	city	Delhi			
	Key	Value	Description		

Body



200 OK



18 ms



286 B



{ } JSON



Preview



1

{

2

"name": "Akshay Kumar\n",

3

"age": "25",

4

"city": "Delhi"

5

}



EXPLORER index.js

FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > ...

```
14 app.get('/', (req, res) => {  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17   
18 app.get('/about', (req, res) => {  
19   res.send(req.hostname)  
20 })  
21   
22   
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



localhost



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.ip)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

I:1



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.ips)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.method)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GET



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > ...

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.originalUrl)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

/about

localhost:3000/about?name=Yahubaba - localhost:3000/about?name=Yahubaba x

localhost:3000/about? - Google Search

localhost:3000/about?





EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.path)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

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localhost:3000/about?name=Yahubaba

Google Lens



/about

EXPLORER ... index.js

FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15     res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19     res.send(req.protocol)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

http





EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15   res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19   res.send(req.secure)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
13 app.get('/', (req, res) => {  
14  
15     res.jsonp({ name: 'YahuBaba', age: 25})  
16 })  
17  
18 app.get('/about', (req, res) => {  
19     res.send(req.route)  
20 })  
21  
22  
23
```

PROBLEMS

OUTPUT

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TERMINAL

PORTS



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > app.get('/about:userid') callback > message

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about/:userid', (req, res) => {
19   if(req.accepts('html')){
20     res.send("<h1>Hello HTML</h1>")
21   }else if(req.accepts('json')){
22     res.send({ message: 'Hello JSON'})
23   }
24 })
25
26 |
27
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > app.get('/about:userid') callback

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about/:userid', (req, res) => {
19   if(req.accepts('html')){
20     res.send("<h1>Hello HTML</h1>")
21   }else if(req.accepts('json')){
22     res.send({ message: 'Hello JSON'})
23   }else if(req.accepts('xml')){
24     res.send("<message>Hello XML</message>")
25   }
26 })
27
28
29
```

I

PROBLEMS

OUTPUT

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EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

index.js > app.get('/about/:userid') callback

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about/:userid', (req, res) => {
19   if(req.accepts('html')){
20     res.send("<h1>Hello HTML</h1>")
21   }else if(req.accepts('json')){
22     res.send({ message: 'Hello JSON'})
23   }else if(req.accepts('xml')){
24     res.send("<message>Hello XML</message>")
25   }else{
26     res.send("Content type not supported.")
27   }
28 })
29
30
31
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



EXPLORER



index.js 1 x



FIRST-PROJECT



> files

> node_modules

views

user.ejs

index.js 1

package-lock.json

package.json



index.js > app.get('/about') callback

15 res.jsonp({ name: 'YahuBaba', age: 25})

16 })

17

18 app.get('/about', (req, res) => {

19 res.send(req.headers.)

20 (property) IncomingHttp × accept-language?

21 pHeaders["accept-langu accept-patch?

22 age"]?: string | undef accept-ranges?

23 ined accept?

accept-control-allow-credentials?

accept-control-allow-headers?

accept-control-allow-methods?

accept-control-allow-origin?

accept-control-expose-headers?

accept-control-max-age?

accept-control-request-headers?

accept-control-request-method?

age?

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
localhost:3000/about/ x + v
localhost:3000/about/
{
  "host": "localhost:3000",
  "connection": "keep-alive",
  "cache-control": "max-age=0",
  "sec-ch-ua": "\"Chromium\";v=\"134\"", "\"Not:A-Brand\";v=\"24\"", "\"Google Chrome\";v=\"134\"",
  "sec-ch-ua-mobile": "?0",
  "sec-ch-ua-platform": "\"Windows\"",
  "upgrade-insecure-requests": "1",
  "user-agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36",
  "accept": "text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8;/signed-exchange;v=b3;q=0.7",
  "sec-fetch-site": "none",
  "sec-fetch-mode": "navigate",
  "sec-fetch-user": "?1",
  "sec-fetch-dest": "document",
  "accept-encoding": "gzip, deflate, br, zstd",
  "accept-language": "en-US,en;q=0.9,la;q=0.8,hi;q=0.7",
  "if-none-match": "W/\"13-JZkA5AmsHThc+9+J8HyGWVe3dDQ\""
}
```



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about', (req, res) => {
19   res.send(req.headers.host)
20 })
21
22
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

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localhost:3000



EXPLORER



index.js



FIRST-PROJECT



> files

> node_modules

views



user.ejs



index.js



package-lock.json



package.json



index.js > app.get('/about') callback

```
15 res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about', (req, res) => {
19   res.send(req.get("Host"))
20 })
21
22
23
```

get(name: "set-cookie"):
string[] | undefined

Return request header.

The **Referrer** header field is
special-cased, both **Referrer** and
Referer are interchangeable.

1/2 Examples:

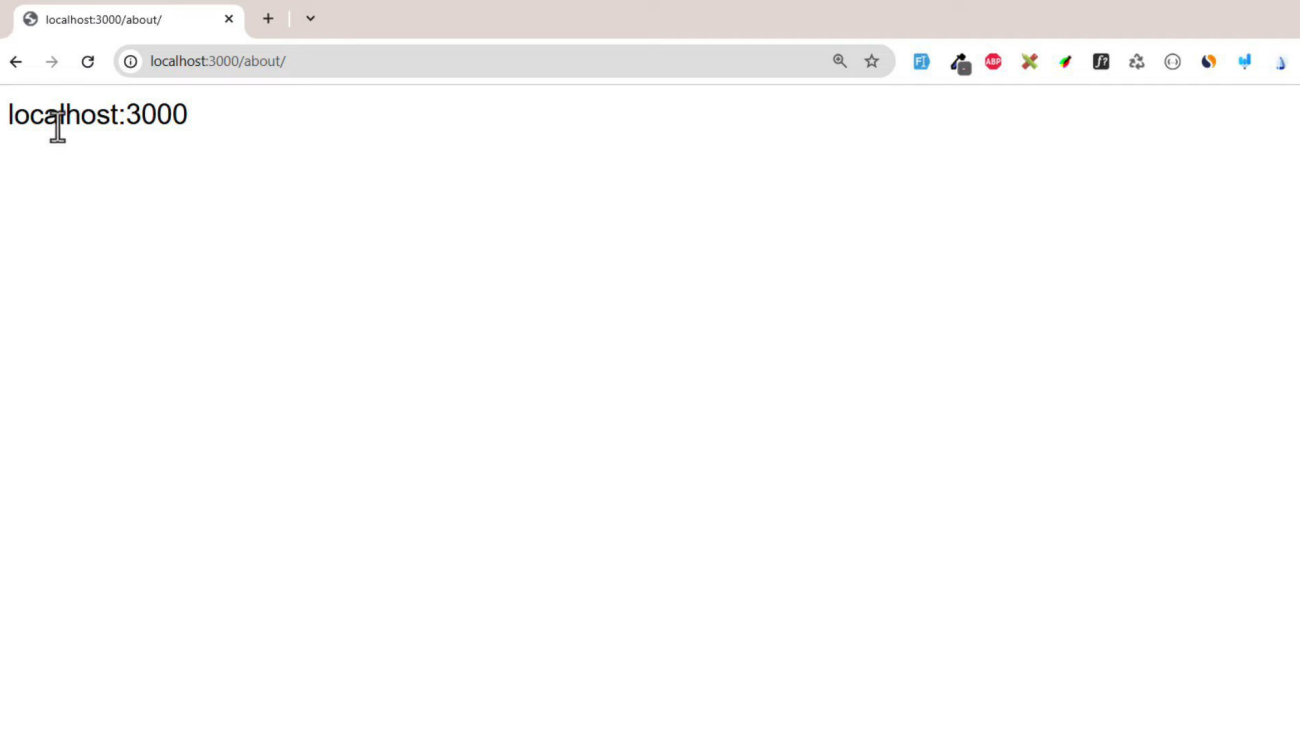
PROBLEMS

OUTPUT

DEBUG CONSOLE

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EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about', (req, res) => {
19   res.send(req.get("Connection"))
20 })
21
22
23
```

PROBLEMS

OUTPUT

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keep-alive



EXPLORER



index.js



FIRST-PROJECT

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json



index.js > app.get('/about') callback

```
15   res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about', (req, res) => {
19   res.send(req.get("Accept"))
20 })
21
22
23
```



PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

EXPLORER index.js

FIRST-PROJECT index.js > app.get('/about') callback

- > files
- > node_modules
- > views
 - user.ejs
 - index.js
- package-lock.json
- package.json

```
15 res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17
18 app.get('/about', (req, res) => {
19   if(req.is('application/json')){
20     res.send("Valid JSON Data")
21   }else if(req.is('text/html')){
22     res.send("HTML Data")
23   }else{
24     res.status(400).send("Unsupported Content-Type.")
25   }
26 })
27
28
29
```

Unsupported Content-Type.

EXPLORER ... index.js

FIRST-PROJECT index.js > ...

> files

> node_modules

views

user.ejs

index.js

package-lock.json

package.json

```
15 res.jsonp({ name: 'YahuBaba', age: 25})
16 })
17 
18 app.post('/about', (req, res) => {
19   if(req.is('application/json')){
20     res.send("Valid JSON Data")
21   }else if(req.is('text/html')){
22     res.send("HTML Data")
23   }else{
24     res.status(400).send("Unsupported Content-Type.")
25   }
26 })
27 
28 
29
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

New Import

Overview

POST localhost:3000/about +

v

No environment v

📋



localhost:3000/about



Save



Share



POST



localhost:3000/about

Send



Body v

Cookies

x-www-form-urlencoded v

	Key	Value	Description	...	Bulk Edit
☒	name	Akshay Kumar	...		
☒	age	25			
☒	city	Delhi			
	Key	Value	Description		

Body v ⌚

400 Bad Request

135 ms

262 B



HTML v

Preview

Visualize



1 Unsupported Content-Type.

New Import

Overview

POST localhost:3000/about

+

v

No environment

v

📄



localhost:3000/about



Save

v

Share



POST

v

localhost:3000/about

Send

v

Body v

raw

v

JSON

v

Cookies

Beautify

Body v ⌚



HTML

v

▶ Preview



Visualize

v

200 OK

19 ms

242 B



...



```
1 {  
2   "name": "Yahubaba",  
3   "age": 25  
4 }
```

```
1 Valid JSON Data
```

New Import

Overview

POST localhost:3000/about +

Dropdown

No environment

Help

HTTP localhost:3000/about

Save

Share

Code

POST

localhost:3000/about

Send

Body

Cookies

Body

200 OK

17 ms

235 B

Globe

More

raw

HTML

Beautify

HTML

Preview

Visualize

Dropdown

Refresh

Copy

Search

Link

```
1 <h1>Hello</h1>
```

```
1 HTML Data
```

I